

LAB 05

Title: String Tokenization Using C++ with Debugging Techniques

Objective

To learn how to debug a C++ program using Breakpoints and the Watch Window, and to understand how a string is divided into individual tokens.

Procedure

1. Complete environment setup.
2. Write String Tokenization code.
3. Set break points

Breakpoint Location	Reason
getline(cin, s);	To verify that the complete input string is read correctly.
if(s[j] == ' ')	To check whether the current character is a space and determine when a token should be separated.
token += s[j];	To observe how each character is appended to the token variable.

4. Execute with Next Line
5. Track the changes on watch

Code with break-points

```

Tokenization.cpp
1  #include <bits/stdc++.h>
2  using namespace std;
3
4  int main()
5  {
6      int n;
7      cin >> n;
8      cin.ignore();
9
10     for(int i = 0; i < n; i++)
11     {
12         string s;
13         getline(cin, s);
14
15         cout << "String " << i+1 << endl;
16
17         string token = "";
18         int count = 1;
19
20         for(int j = 0; j < s.length(); j++)
21         {
22             if(s[j] == ' ')
23             {
24                 cout << "Token " << count++ << " " << token << endl;
25                 token = "";

```

```

26     }
27     else
28     {
29         token += s[j];
30     }
31 }
32
33 if(token != "")
34     cout << "Token " << count++ << " " << token << endl;
35
36 cout << endl;
37 }
38
39 return 0;
40 }

```

Output

The screenshot shows the IDE with the following state:

- Project:** Tokenization.cpp
- Debug Console:**
 - n = 2
 - i = 0
 - j = 28
 - s = "Hello My Name is Himel Sarder"
 - count = 7
 - token = "Sarder"
- Code Editor:** Line 36 is highlighted, showing `cout << endl;`.
- Output Window:**

```

2
Hello My Name is Himel Sarder
String 1
Token 1 Hello
Token 2 My
Token 3 Name
Token 4 is
Token 5 Himel
Token 6 Sarder

```

The screenshot shows the IDE with the following state:

- Project:** Tokenization.cpp
- Debug Console:**
 - n = 2
 - i = 1
 - j = Not found in current context
 - s = "How Are You"
 - count = 4
 - token = "You"
- Code Editor:** Line 40 is highlighted, showing `return 0;`.
- Output Window:**

```

2
Hello My Name is Himel Sarder
String 1
Token 1 Hello
Token 2 My
Token 3 Name
Token 4 is
Token 5 Himel
Token 6 Sarder

How Are You
String 2
Token 1 How
Token 2 Are
Token 3 You

```